

# COMPASS: TEMPEST Discrete DOC Data QAQC

August 2026

2026-10-07

## Run Information

```
#identify which section you are in  
cat("Run Information")
```

```
## Run Information
```

```
#a link to the Gitbook or whatever protocol you are using for this analysis  
#steph will add this soon
```

```
#anything that needs to be changed do this in the first chunk
```

```
Date_Run = "08/15/26"
```

```
Run_by = "Stephanie J. Wilson"
```

```
Script_run_by = "Stephanie J. Wilson"
```

```
run_notes = "Check standard getting old, C standards coming up a bit higher than 10% of the expected"
```

```
#file path and name for summary file
```

```
raw_file_name = "tmp_doc_raw_data_2026/TMP_202608.txt"
```

```
#file path and name for the all peaks file
```

```
raw_allpeaks_name = "tmp_doc_raw_data_2026/TMP_202608_allpeaks.txt"
```

```
#file path and name for processed data after QAQC
```

```
processed_file_name = "tmp_doc_processed_data_2026/TMP_PW_DOC_Processed_202608.csv"
```

```
#check standard concentrations - Update if running different checks:
```

```
chk_std_c = 1
```

```
chk_std_n = 1
```

```
#Log path
```

```
Log_path = "tmp_doc_raw_data_2026/COMPASS_TMP_TOCTN_QAQClog_2026.csv"
```

## Setup

```
##Pull in active porewater inventory
```

## Import Data Functions

### Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 4
```

| ##   | sample_name                 | npoc_raw | tdn_raw | run_datetime          |
|------|-----------------------------|----------|---------|-----------------------|
| ##   | <chr>                       | <dbl>    | <dbl>   | <chr>                 |
| ## 1 | TMP_C_C6_50cm_20260813      | 3.85     | 1.72    | 8/19/2026 11:23:56 PM |
| ## 2 | TMP_FW_C6_15cm_20260810     | 11.6     | 0.493   | 8/19/2026 11:55:58 PM |
| ## 3 | TMP_FW_C6_30cm_20260810     | 4.05     | 0.253   | 8/20/2026 12:22:23 AM |
| ## 4 | TMP_FW_C6_30cm_20260810_dup | 4.10     | 0.239   | 8/20/2026 12:52:39 AM |
| ## 5 | TMP_FW_C6_50cm_20260810     | 2.20     | 0.148   | 8/20/2026 1:14:38 AM  |
| ## 6 | TMP_FW_D5_15cm_20260810     | 15.4     | 0.559   | 8/20/2026 1:48:43 AM  |

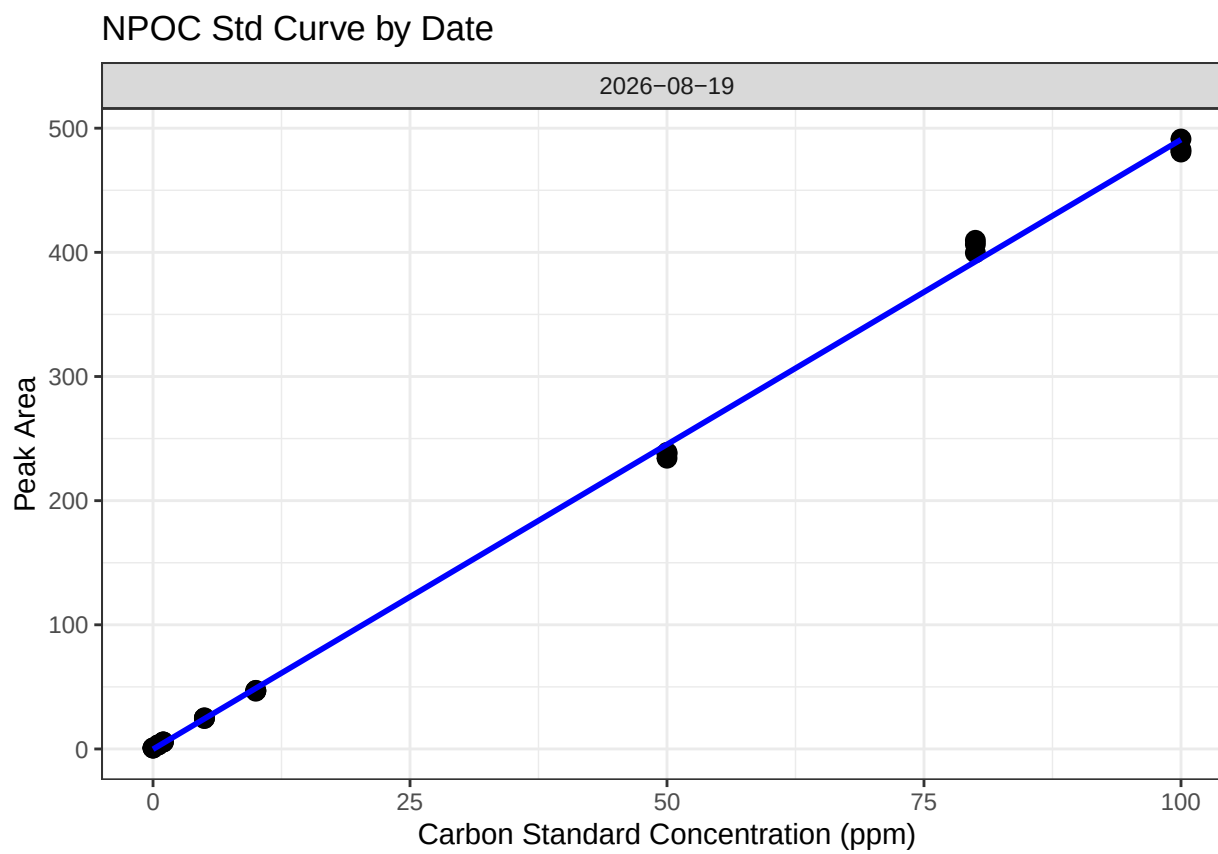
## Assessing standard Curves

```
## Assess the Standard Curve
```

```
## New names:
```

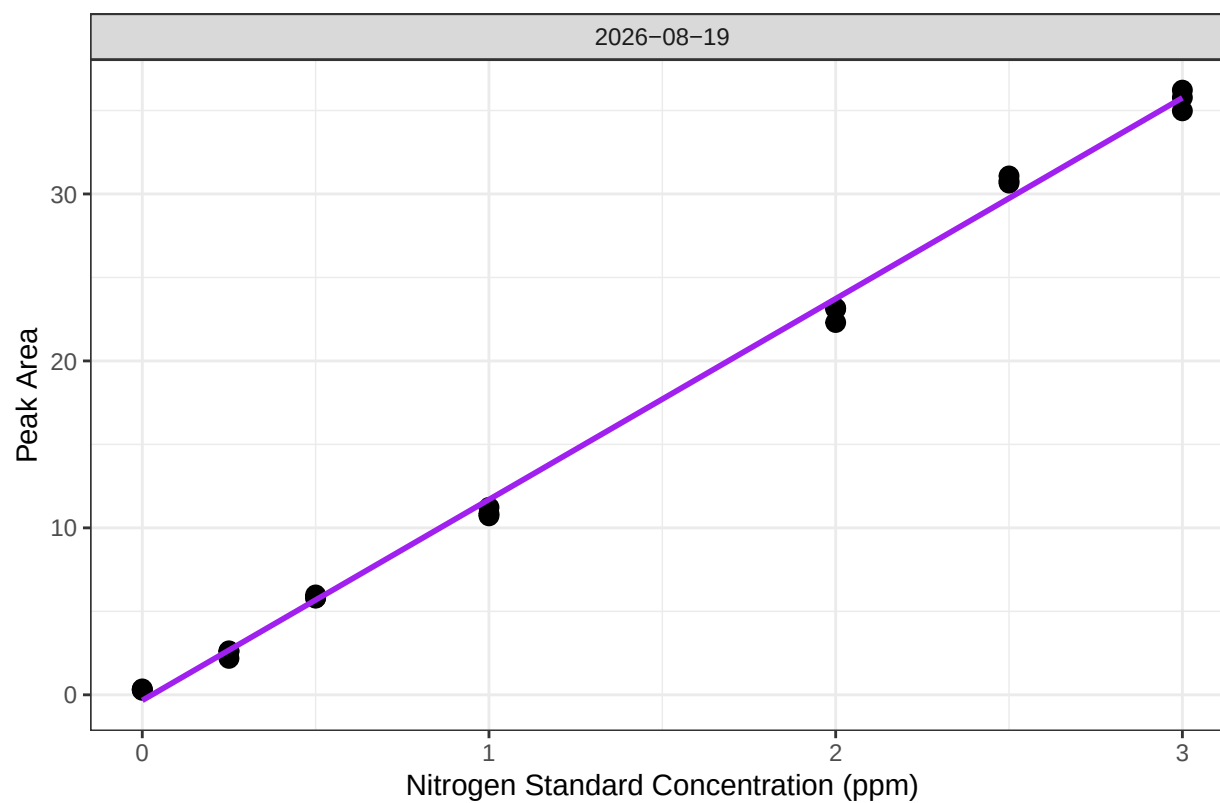
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## * '' -> '...18'
```

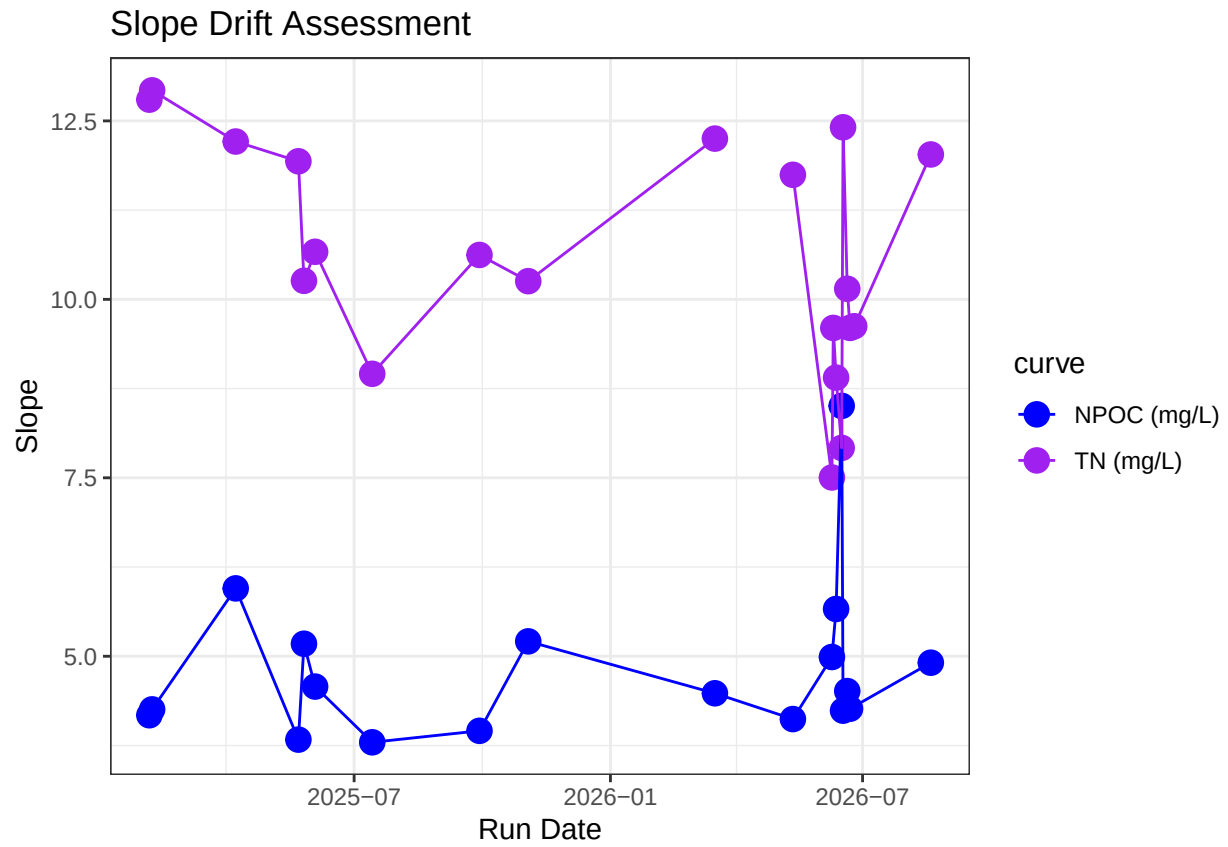


```
## 'geom_smooth()' using formula = 'y ~ x'
```

TN Std Curve by Date



```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
## [1] "NPOC Curve r2 GOOD"
```

```
## [1] "TN Curve r2 GOOD"
```

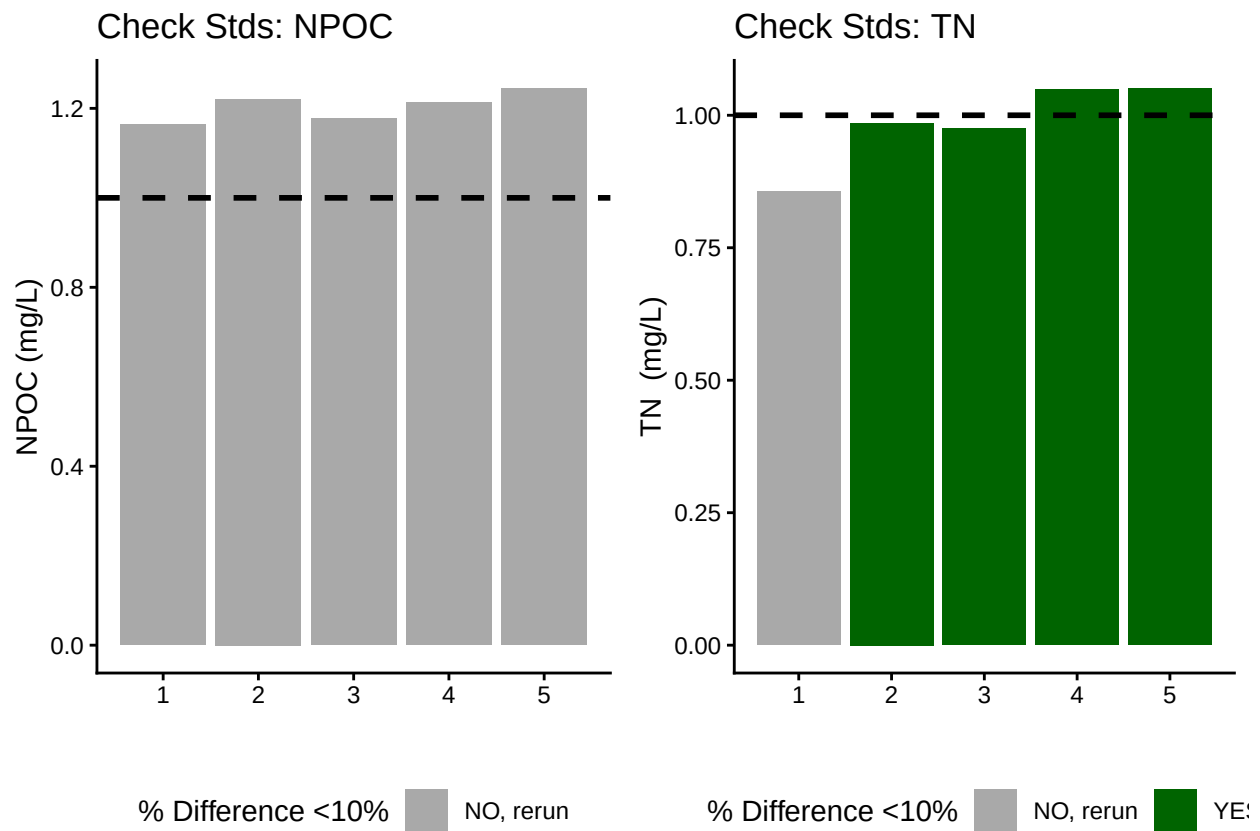
## Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] "Carbon Check Standard RSD within Range"
```

```
## [1] "Nitrogen Check Standard RSD within Range"
```



```
## [1] "<60% of Carbon Check Standards are within range of the expected concentration - REASSESS"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of the expected concentration"
```

## Assess Blanks

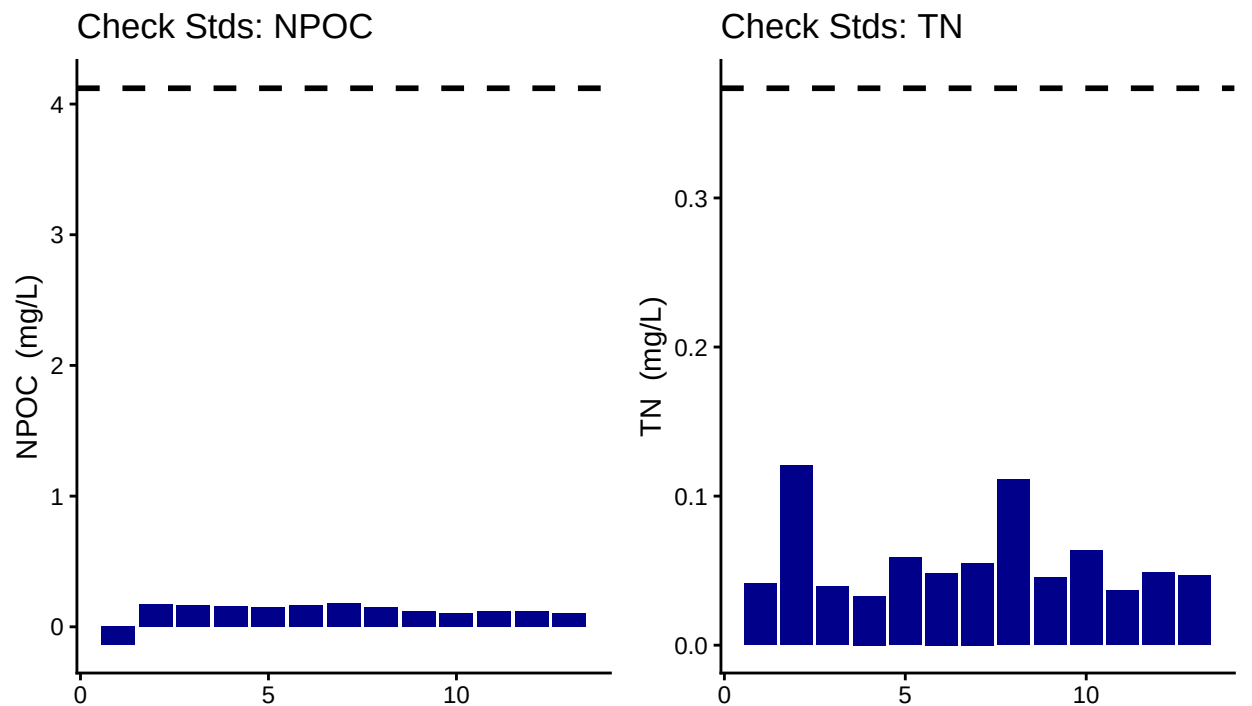
```
## Assess Blanks
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are below the lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are below the lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ☒ YE

Blank Conc <25% Quartile Samples ☒ YE

```
## carbon blanks:
```

```
## [1] 0.1195154
```

```
## nitrogen blanks:
```

```
## [1] 0.05757
```

## Assess Duplicates

```
## Assess Duplicates
```

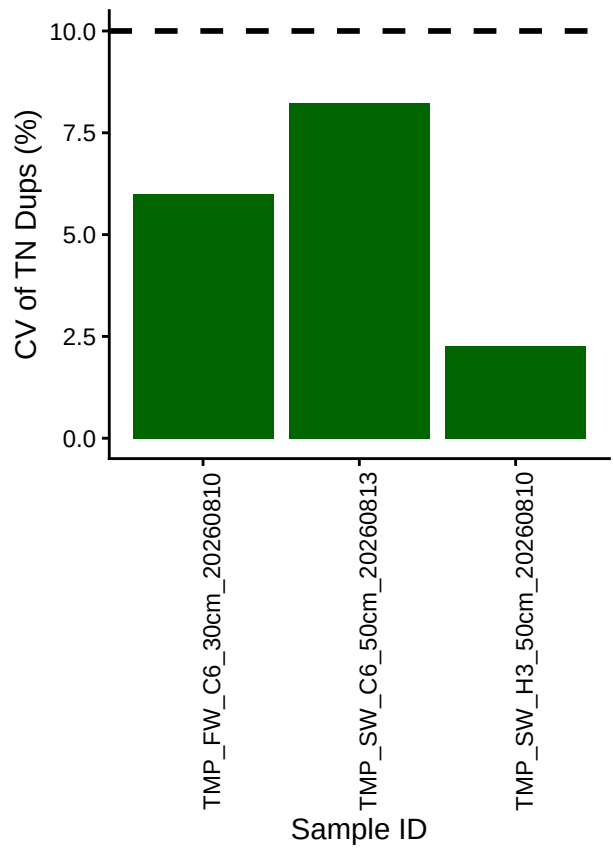
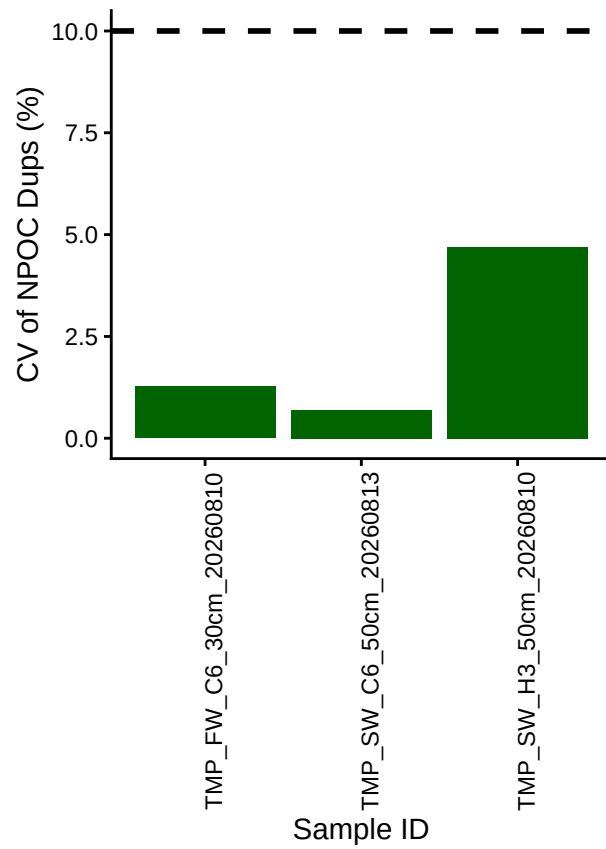
```
## # A tibble: 3 x 3
```

```
##   sample_name      npoc_raw_dup tdn_raw_dup
##   <chr>          <dbl>          <dbl>
## 1 TMP_FW_C6_30cm_20260810      4.10      0.239
## 2 TMP_SW_C6_50cm_20260813      5.51      0.884
## 3 TMP_SW_H3_50cm_20260810      5.87      1.22
```

```
##           sample_name npoc_raw tdn_raw      run_datetime
## 1 TMP_FW_C6_30cm_20260810    4.054 0.2534 8/20/2026 12:22:23 AM
## 2 TMP_SW_C6_50cm_20260813    5.545 0.9571 8/20/2026 7:46:31 AM
## 3 TMP_SW_H3_50cm_20260810    5.610 1.1930 8/20/2026 11:34:26 AM
##           npoc_flag tdn_flag npoc_raw_dup tdn_raw_dup
## 1 NPOC checks out of range      4.103      0.2392
## 2 NPOC checks out of range      5.509      0.8837
## 3 NPOC checks out of range      5.868      1.2190
```

```
##           sample_name npoc_raw tdn_raw      run_datetime
## 1 TMP_FW_C6_30cm_20260810    4.054 0.2534 8/20/2026 12:22:23 AM
## 2 TMP_SW_C6_50cm_20260813    5.545 0.9571 8/20/2026 7:46:31 AM
## 3 TMP_SW_H3_50cm_20260810    5.610 1.1930 8/20/2026 11:34:26 AM
##           npoc_flag tdn_flag npoc_raw_dup tdn_raw_dup npoc_dups_cv
## 1 NPOC checks out of range      4.103      0.2392      1.2689107
## 2 NPOC checks out of range      5.509      0.8837      0.6892715
## 3 NPOC checks out of range      5.868      1.2190      4.6936561
##   npoc_dups_cv_flag tdn_dups_cv tdn_dups_cv_flag
## 1                YES      5.992896                YES
## 2                YES      8.226596                YES
## 3                YES      2.269366                YES
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

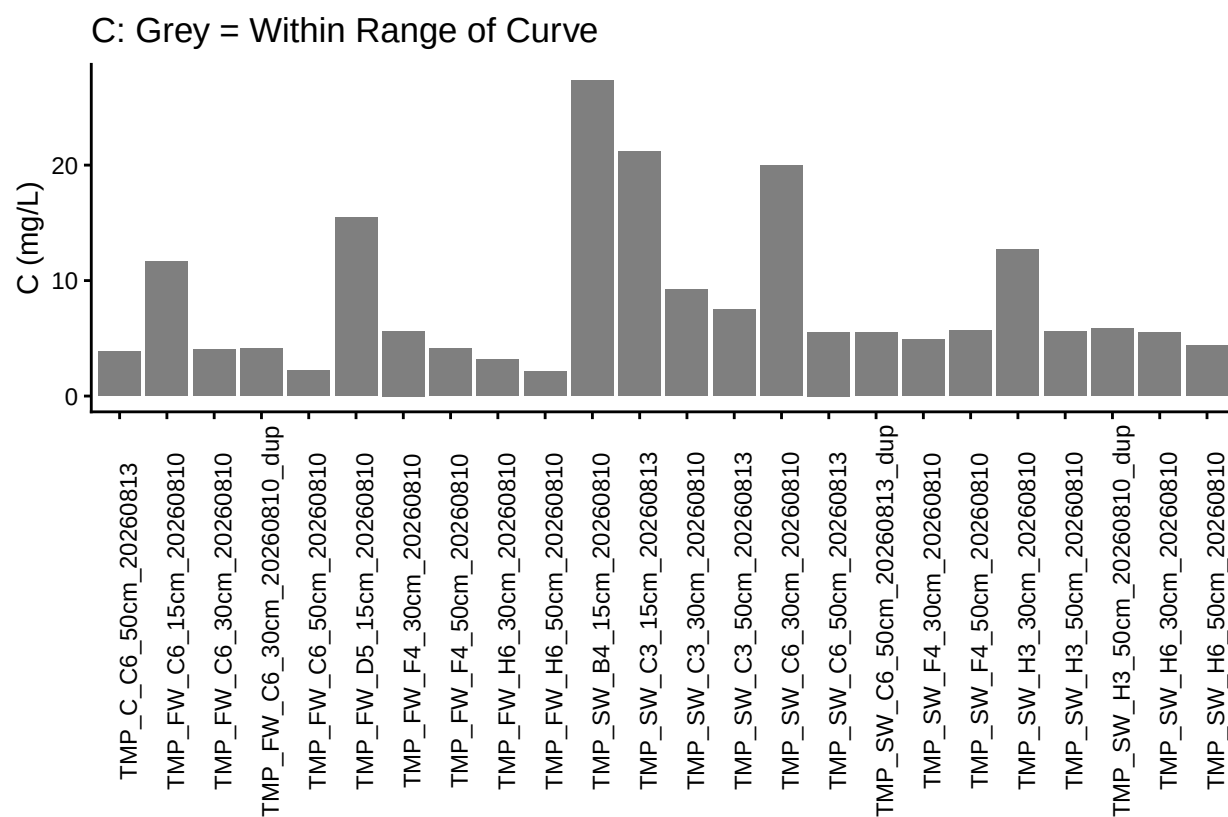


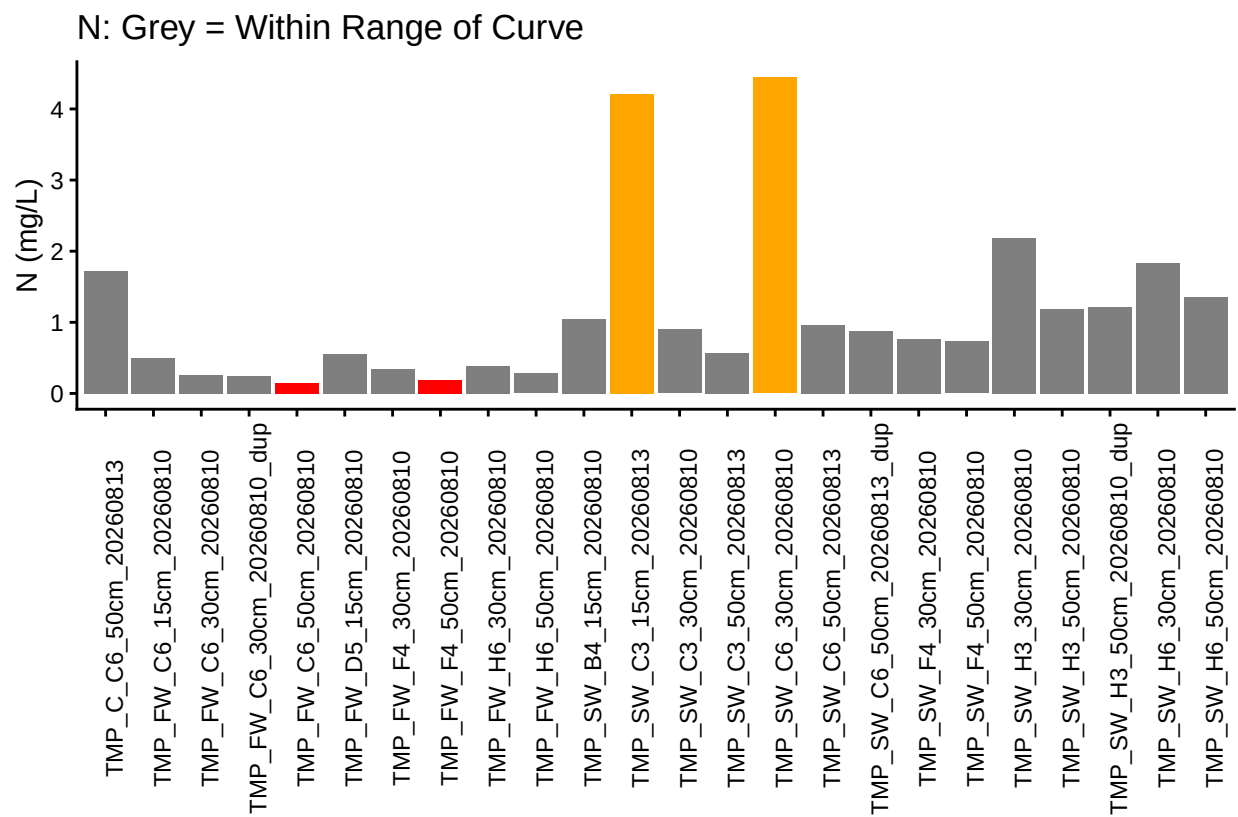
```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

## Sample Flagging

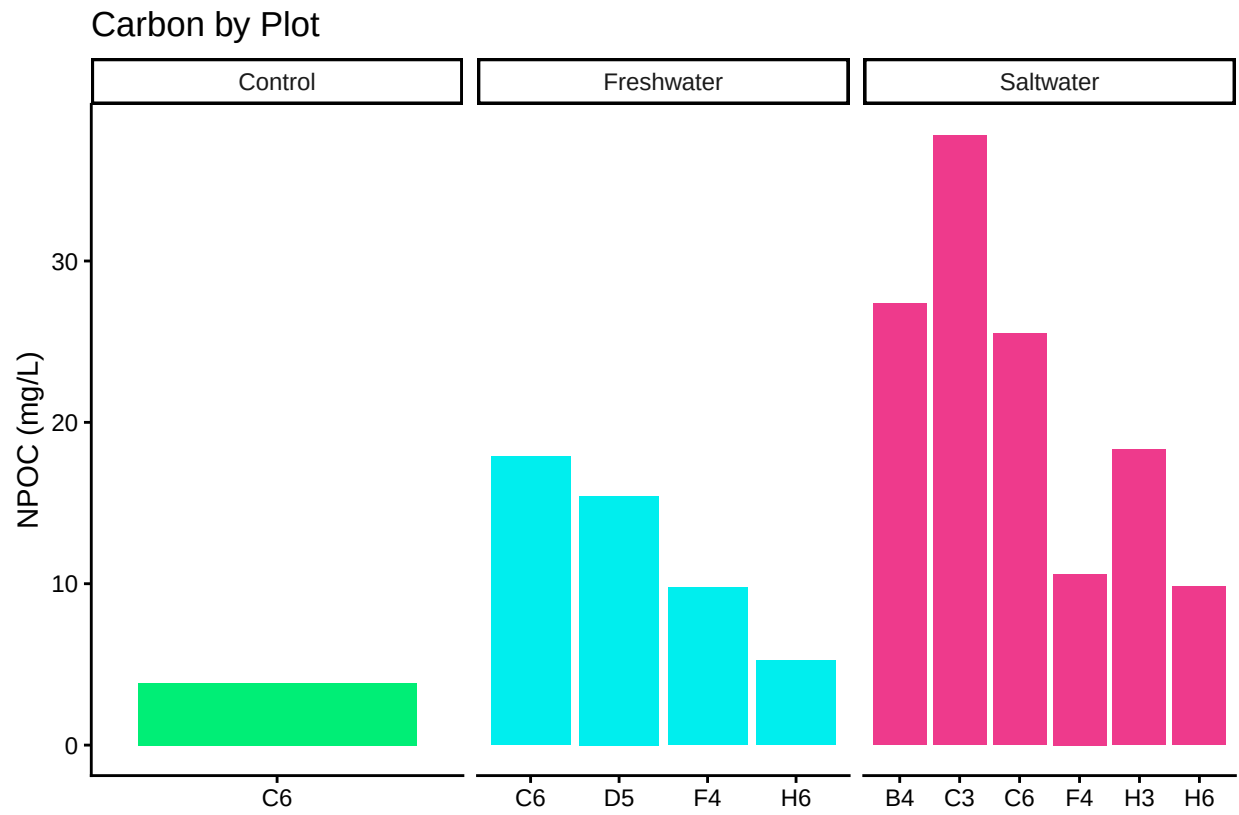
## Sample Flagging

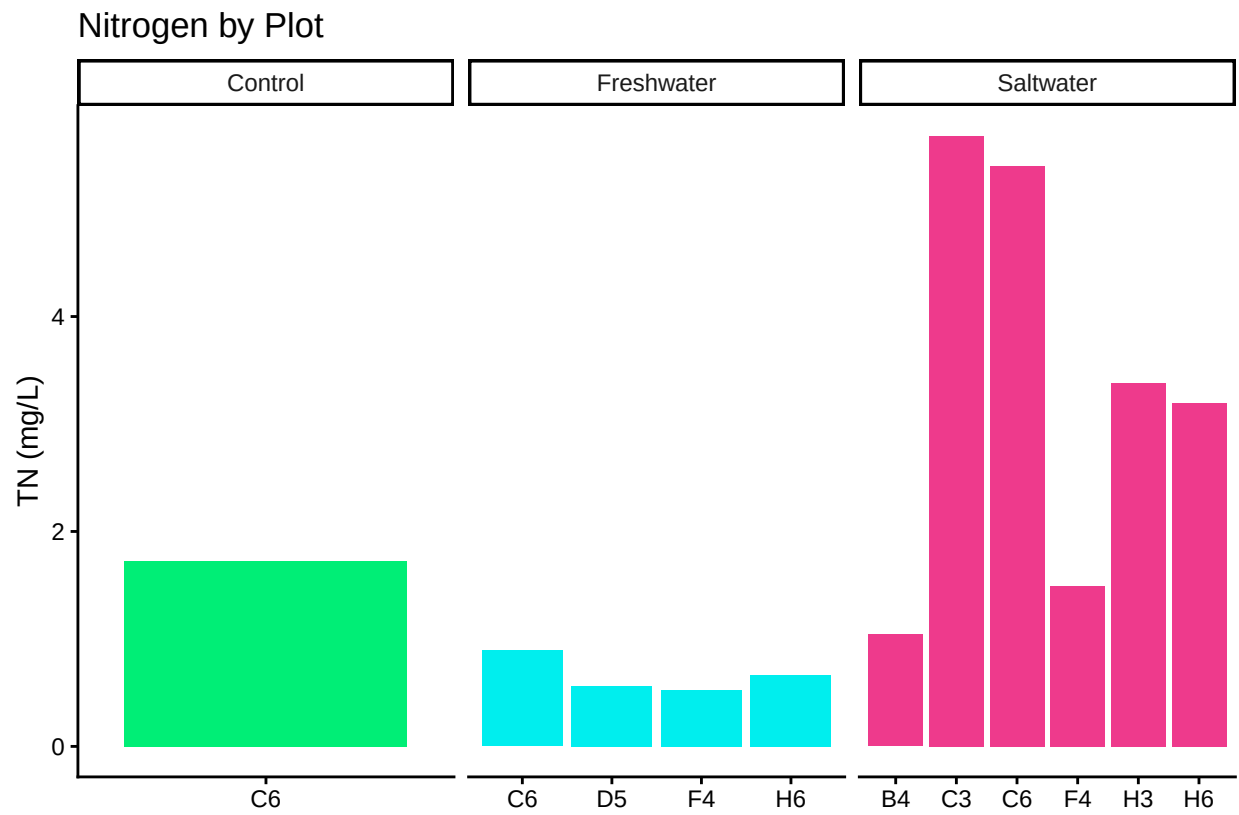




## Visualize Data by Plot

## Visualize Data





## Convert data from mg/L to uMoles/L

### Add in/check metadata

```
## Check Sample IDs with Metadata
```

```
## # A tibble: 21 x 2
##   sample_name      raw_metadata
##   <chr>          <lgl>
## 1 TMP_C_C6_50cm_20260813 TRUE
## 2 TMP_FW_C6_15cm_20260810 TRUE
## 3 TMP_FW_C6_30cm_20260810 TRUE
## 4 TMP_FW_C6_50cm_20260810 TRUE
## 5 TMP_FW_D5_15cm_20260810 TRUE
## 6 TMP_FW_F4_30cm_20260810 TRUE
## 7 TMP_FW_F4_50cm_20260810 TRUE
## 8 TMP_FW_H6_30cm_20260810 TRUE
## 9 TMP_FW_H6_50cm_20260810 TRUE
## 10 TMP_SW_B4_15cm_20260810 TRUE
## # i 11 more rows
```

### Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 22
##   Project      Plot Grid Depth_cm sample_type Vial_ID date npoc_mgL npoc_uM
##   <chr>      <chr> <chr> <chr>    <chr>      <chr> <chr>    <dbl>  <dbl>
## 1 COMPASS: TEMP~ Cont~ C6    50cm    DOC        C_C6_5~ 2026~    3.85   321.
## 2 COMPASS: TEMP~ Fres~ C6    15cm    DOC        FW_C6_~ 2026~    11.6   971.
## 3 COMPASS: TEMP~ Fres~ C6    30cm    DOC        FW_C6_~ 2026~    4.05   338.
## 4 COMPASS: TEMP~ Fres~ C6    50cm    DOC        FW_C6_~ 2026~    2.20   183.
## 5 COMPASS: TEMP~ Fres~ D5    15cm    DOC        FW_D5_~ 2026~    15.4  1287.
## 6 COMPASS: TEMP~ Fres~ F4    30cm    DOC        FW_F4_~ 2026~    5.63   469.
## # i 13 more variables: npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>,
## #   Evacuation_date_YYYYMMDD <dbl>, Collection_Date_YYYYMMDD <dbl>,
## #   Collection_Start_Time_24hrs <dbl>, Collection_End_Time_24hrs <dbl>,
## #   EST_EDT <chr>, Volume_mL <dbl>
```

```
#end
```